

HAT-P-37b

R-2506

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_150604 · created 2026-08-24T23:47:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457177.93799 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.019
Transit depth (Rp/Rs)^2	2.11 +/- 0.57 [%]
Orbital Inclination (inc)	89.61 +/- 1.4
Airmass coefficient 1 (a1)	1.579 +/- 0.035
Airmass coefficient 2 (a2)	-0.43 +/- 0.16
Scatter in the residuals of the lightcurve fit is	2.21 %
Best Comparison Star	#3 - [45, 297]
Optimal Aperture	2.92
Optimal Annulus	12.91
Transit Duration (day)	0.0873 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.145 $\pm$ 0.019 vs 0.1394 (deviation 0.27 σ)
Duration fit vs published	0.0873 d vs 0.0983 d (deviation 5.33 σ)
Mid-time O-C	-2.0 min
Depth SNR	7.0
Residual scatter	2.21 %

Light curve

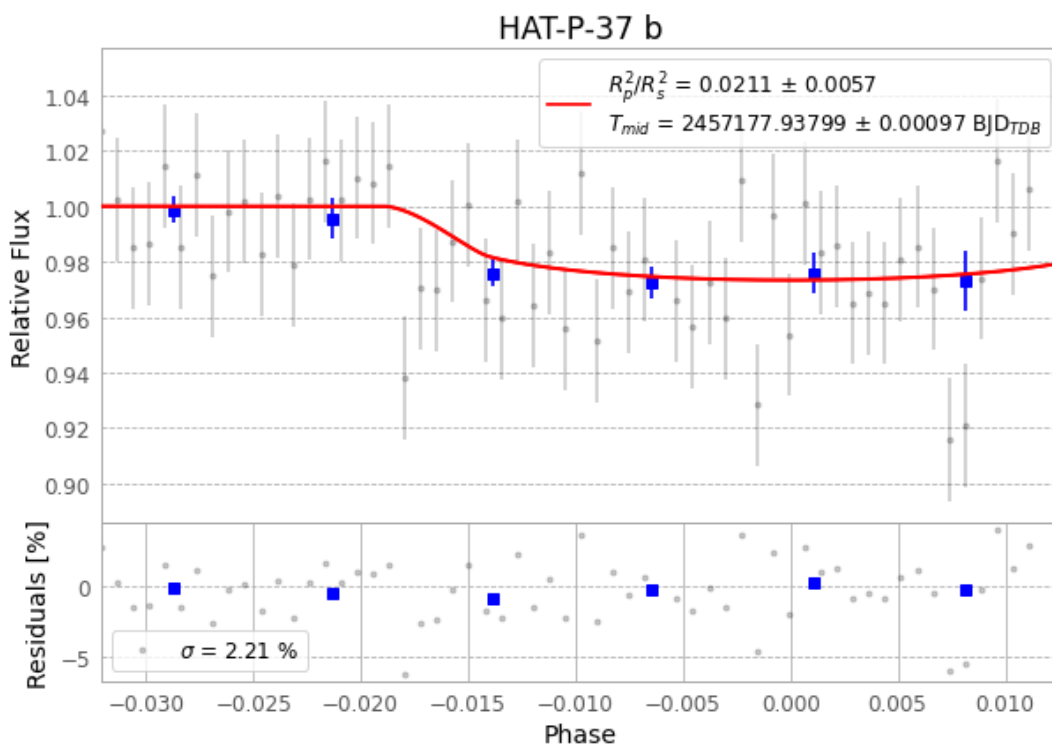


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-06-04.png (SHA-256 8c1f50f55d9a3952...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [175, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a5861ac86022cef9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), HATP-37150604081812.FITS → HATP-37150604111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150604.log (33a50c93f7f3...), FinalLightCurve_HAT-P-37 b_2015-06-04.png (8c1f50f55d9a...), FinalLightCurve_HAT-P-37 b_2015-06-04.csv (0bc09646f3da...)

Compute resources

Wall time 343.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:47Z.